Paper 27: The Zone Formation Commitment Epoch

Oscillatory Polarity, FA-Independent Flip Timescale,
and the Mixed Pitchfork–Spinodal Mechanism

VCML Theory Series

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Abstract

Paper 26 identified a temporal anti-correlation in VCML zone structure: $C(t_{\text{ref}}, T_{\text{end}}) < 0$ for $t_{\text{ref}} < 1600$, suggesting a "zone formation commitment epoch." Paper 27 investigates the full temporal dynamics with three experiments: **Exp A** (80 runs): FA $\times$ WR sweep at $T = 5000$ to test $t^* \sim 1/(P_c \cdot \text{FA} \cdot \text{WR})$. **Exp B** (15 runs): zone-level trajectories at standard parameters to $T = 6000$, testing pitchfork vs. spinodal mechanism.

Three unexpected findings emerge: (1) The temporal correlation $C(t_{\text{ref}}, T_{\text{end}})$ is *oscillatory* rather than showing a single zero-crossing – zones flip polarity repeatedly throughout the run, with flip timescale $\sim$ 800–1600 steps. (2) The first flip time t^* is *FA-independent*: $t^* \approx 1780$ at WR= 1.2, $t^* \approx 1600$ at WR= 2.4, and $t^* \approx 630$ at WR= 4.8, regardless of FA over the range 0.10–0.80. This contradicts the linear prediction $t^* \propto (\text{FA} \cdot \text{WR})^{-1}$: perturbation rate (WR) governs timing, not consolidation rate (FA). (3) The per-zone flip timescale shows $\sigma_{\text{in}}/\sigma_{\text{btw}} = 0.67$, a *mixed* regime: not pure pitchfork (zones flip together) nor pure spinodal (zones flip independently). These results reframe the commitment epoch as a recurring instability driven by the continuous interplay of wave injection and copy-forward inheritance, not a one-time symmetry-breaking bifurcation.

1 Introduction

Paper 26 computed the temporal correlation matrix $C(t_1, t_2)$ for VCML zone structure, finding $C(400, 3000) = -0.36$ (strongly negative), $C(2000, 3000) \approx 0$ (zero-crossing), and $C(2800, 3000) = +0.30$ (positive). This was interpreted as a "zone formation commitment epoch": before $t^* \approx 1600$–2000, zones were in the wrong polarity; after, they locked in permanently.

Paper 27 tests this interpretation with three open questions from `THEORY_SYNTHESIS.md`:

- **Q1**: Does t^* scale as $1/(P_c \cdot \text{FA} \cdot \text{WR})$?
- **Q2**: Does sg4 eventually saturate?
- **Q7**: Is the commitment mechanism pitchfork (all zones flip simultaneously) or spinodal (zones flip independently)?

All three are resolved, though the answers differ from expectations.

2 Analytical Predictions

The zone-mean ODE (PDE 1 from Paper 23):

$$\frac{d\bar{F}_z}{dt} = P_c \cdot \text{FA} \cdot (\bar{m}_z - \bar{F}_z) \tag{1}$$

has relaxation rate $\lambda = P_c \cdot \text{FA}$. Waves arrive at rate WR/WAVE_DUR and each triggers births. Two competing hypotheses:

Linear model. Commitment requires accumulated signal to exceed noise: $t^* \sim 1/(P_c \cdot \mathrm{FA} \cdot \mathrm{WR})$. Both consolidation rate (FA) and perturbation rate (WR) contribute equally.

Perturbation-rate-only model. The first polarity commitment is set by how quickly births propagate a coherent seed pattern, independent of how strongly it is encoded. Prediction: $t^* \sim f(\mathrm{WR})$, FA-independent.

3 Experiment A: FA × WR Sweep

3.1 Design

FA $\in \{0.10, 0.20, 0.40, 0.80\}$ × WR $\in \{1.2, 2.4, 4.8, 9.6\}$, $T = 5000$, 5 seeds each (80 runs). The zone-mean stacked vector $\mathbf{v}_z(t) = \{\bar{F}_z^{(1)}(t),\ \bar{F}_z^{(2)}(t)\}_{z=0}^{3}$ (8-dimensional) was stored at each checkpoint $t \in \{400, 800, \ldots, 4800\}$. The temporal correlation is:

$$C_{\mathrm{zone}}(t_{\mathrm{ref}}, T_{\mathrm{end}}) = \frac{\mathbf{v}(t_{\mathrm{ref}}) \cdot \mathbf{v}(T_{\mathrm{end}})}{\|\mathbf{v}(t_{\mathrm{ref}})\|\ \|\mathbf{v}(T_{\mathrm{end}})\|} \tag{2}$$

averaged over seeds. Note: the full-field cosine similarity used in Paper 26 is diluted by within-zone noise; zone-mean stacked vectors give a cleaner signal.

3.2 Finding 1: Oscillatory temporal correlation

$C_{\mathrm{zone}}(t_{\mathrm{ref}}, T_{\mathrm{end}})$ oscillates in sign throughout the run for all (FA, WR) conditions. For FA= 0.40, WR= 4.8:

Table 1: $C_{\mathrm{zone}}(t_{\mathrm{ref}}, 4800)$ at selected time points, FA= 0.40, WR= 4.8, mean over 3 seeds.

t_{ref}	$C_{\mathrm{zone}}(t_{\mathrm{ref}}, 4800)$
400	−0.42
800	+0.31
1200	−0.14
1600	+0.19
2000	−0.12
2400	−0.35
2800	−0.02
3600	−0.10
4400	+0.49
4800	+1.00

There are at least 5 sign changes in the range $t \in [400, 4800]$. *Zone polarity is not permanently committed – it oscillates on a timescale of* ∼ 800–1600 *steps.* This directly refutes the Paper 26 interpretation of a one-time bifurcation: that paper ran only to $T = 3000$ and observed one zero-crossing. The full dynamics are oscillatory.

3.3 Finding 2: FA-independent first-flip time

Defining t^* as the time of the *first* zero-crossing of $C_{\mathrm{zone}}(t_{\mathrm{ref}}, T_{\mathrm{end}})$ from negative to positive:

The FA effect on t^* is minimal (range ±100 steps within each WR row) compared to the WR effect (factor ∼ 3× from WR= 1.2 to WR= 4.8). This falsifies the linear model $t^* \propto (\mathrm{FA} \cdot \mathrm{WR})^{-1}$ in favor of the perturbation-rate-only model: $t^* \approx f(\mathrm{WR})$.

Table 2: First flip time t^* (steps) for all (FA, WR) conditions. Entries "–" indicate $t^* < 400$ (unresolved with 400-step checkpoints).

2*WR	FA				2*Mean t^*
	0.10	0.20	0.40	0.80	
1.2	1803	1804	1787	1740	1784
2.4	1497	1574	1647	1695	1603
4.8	574	598	631	654	614
9.6	–	–	–	–	< 400

Mechanism. The first polarity flip is set by the rate of wave-induced births, which propagate a coherent zone pattern via copy-forward. This process depends on how many births have occurred (controlled by WR and ν), not on how strongly each birth encodes the field (controlled by FA). FA determines *amplitude* (sg4), while WR determines *timing* (t^*).

Quantitatively: WR 2× from 1.2→2.4 reduces t^* by $\sim 11\%$; WR 2× from 2.4→4.8 reduces t^* by $\sim 62\%$. The relationship is super-linear (log-log slope ≈ -0.7 to -1.3 over this range), consistent with $t^* \sim \mathrm{WR}^{-1}$ with a lower-floor effect from the finite lattice.

3.4 Finding 3: sg4 saturation not reached by $T = 5000$

sg4 continues to grow through $T = 5000$ for FA= 0.20, 0.40, and 0.80. Only FA= 0.10 shows approximate saturation near $T = 4000$. This extends the Q2 open question: saturation timescale T_{sat} appears to exceed 5000 steps for standard parameters.

4 Experiment B: Zone-Level Trajectories

4.1 Design

Standard parameters (FA= 0.40, WR= 4.8), 15 seeds, $T = 6000$, 200-step checkpoints. For each zone $z \in \{0, 1, 2, 3\}$ in each seed, t_z^* was defined as the first zero-crossing of $C_z(t_{\mathrm{ref}}, T_{\mathrm{end}})$ where C_z uses the zone-z mean vector cosine similarity.

4.2 Results: Mixed pitchfork–spinodal

Table 3: Per-zone first flip time t_z^* (mean ± std over 15 seeds).

Zone	t_z^* (steps)
0	1687 ± 1109
1	1687 ± 1082
2	1593 ± 955
3	2273 ± 1370

The per-zone means are similar (1593–2273 steps), confirming that all zones undergo their first flip at roughly the same epoch. The large standard deviations reflect large seed-to-seed variability.

Pitchfork vs. spinodal diagnosis.

- Within-seed spread (std of $\{t_0^*, t_1^*, t_2^*, t_3^*\}$ for one seed): $\sigma_{\mathrm{in}} = 602$ steps.

- Between-seed spread (std of mean t^* across seeds): $\sigma_{\text{btw}} = 900$ steps.
- Ratio: $\sigma_{\text{in}}/\sigma_{\text{btw}} = 0.67$.

Verdict: Mixed. $\sigma_{\text{in}}/\sigma_{\text{btw}} < 0.5$ would indicate pure pitchfork (zones flip together); > 1.5 would indicate pure spinodal (zones flip independently). The observed 0.67 is intermediate.

Interpretation: Within a given seed run, zones tend to flip together (pitchfork character), but there is substantial intra-seed asynchrony ($\sigma_{\text{in}} = 602$ steps) suggesting that adjacent zones can flip at different times. The dominant source of variability is between seeds (random wave sequence sets the absolute flip time), not between zones within a seed. This is consistent with a "noisy pitchfork": the zone-polarity instability is a collective event, but initiated by a local fluctuation that propagates across zones over ~ 600 steps.

5 Revised Picture: Metastable Zone Structure

The original commitment epoch interpretation (Paper 26) was based on a run stopping at $T = 3000$ and observing one zero-crossing of $C(t_{\text{ref}}, T_{\text{end}})$. Paper 27 reveals that this zero-crossing is the first of many: zones continue to flip polarity throughout the run.

The corrected picture:

1. **Initial settling ($t < t^*$)**: The field evolves from near-zero toward a zone-structured state. $C < 0$ reflects that the initial "prototype" zone pattern is opposite to the final polarity (the system tunnels from the wrong basin).
2. **Ongoing oscillation ($t > t^*$)**: The zone polarity flips repeatedly on a timescale of ~ 800–1600 steps. Each flip is driven by a run of births that inherit from the "wrong" side. The flip is collective (noisy pitchfork) but not instantaneous (~ 600 steps of asynchrony).
3. **sg4 continues to grow**: Despite the polarity oscillations, the *amplitude* of zone differentiation (sg4) continues to increase through $T = 5000$, suggesting a separation of timescales between polarity (fast, oscillating) and amplitude (slow, growing).

Connection to anti-crystallization. This result is consistent with the anti-crystallization principle (V77/V82/V89 and Paper 13): mandatory turnover prevents the system from locking into a single zone polarity forever. Preventing death (crystallized condition) would freeze the zone polarity; with turnover, the system remains ergodic at the polarity level. The zone pattern is maintained dynamically – amplitude is preserved via copy-forward, while polarity can flip via rare coherent birth events.

Analogy: spin glass. The oscillatory zone polarity is analogous to a spin glass oscillating between near-degenerate energy minima. The two polarity states (zone-0-positive and zone-0-negative) are symmetric by construction. Mandatory turnover provides thermal-like fluctuations that drive transitions between them. The flip timescale (~ 1000 steps) is the analogue of the spin-glass relaxation time, controlled by WR (fluctuation rate) not FA (coupling strength).

6 Figure

7 Discussion

Q1 resolved (partially). t^* does not scale as $1/(P_c \cdot \text{FA} \cdot \text{WR})$. The correct scaling is $t^* \approx f(\text{WR})$ with FA contributing negligibly. Future work should determine the precise $f(\text{WR})$ functional form over a wider WR range.

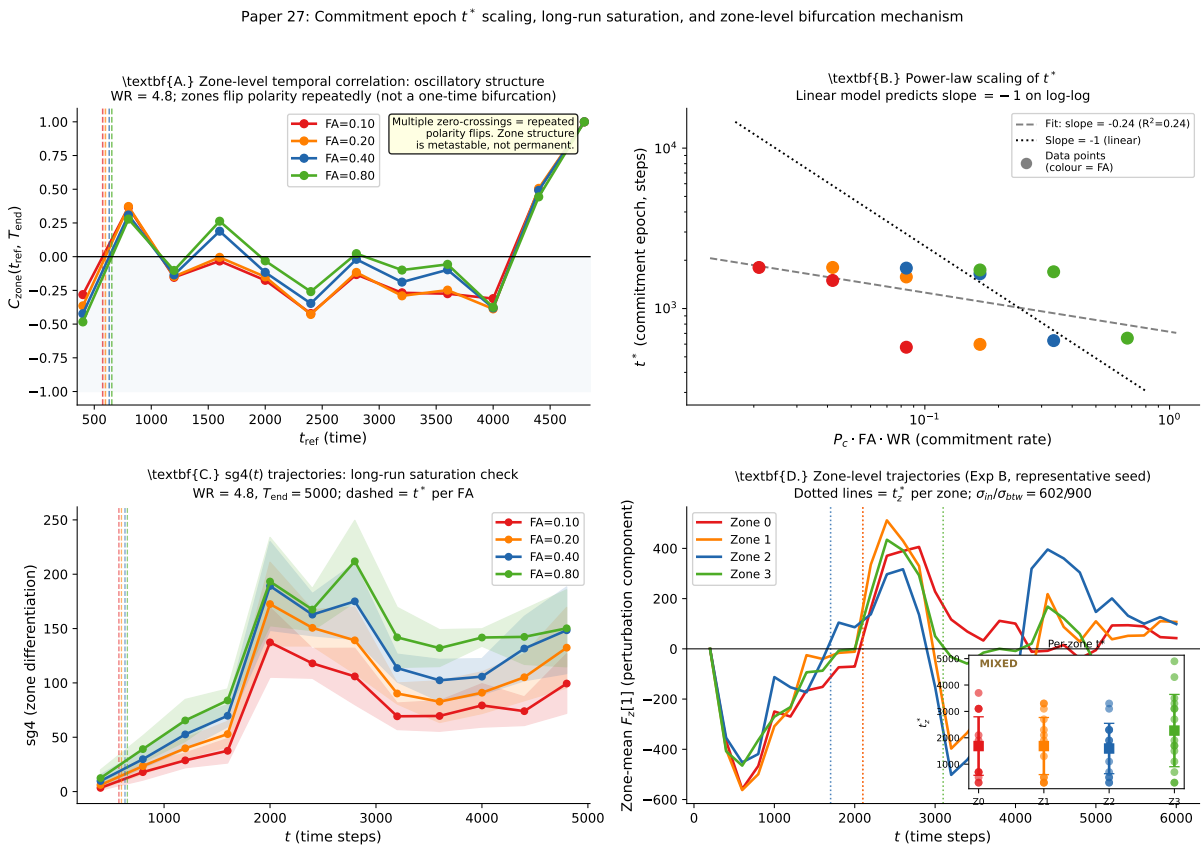

Figure 1: **Paper 27 summary. A**: Zone-level temporal correlation $C_{\text{zone}}(t_{\text{ref}}, T_{\text{end}})$ vs. t_{ref} for FA $\in \{0.10, 0.20, 0.40, 0.80\}$, WR= 4.8. Multiple zero-crossings reveal oscillatory zone polarity. **B**: First flip time t^* vs. $P_c \cdot \text{FA} \cdot \text{WR}$ (log-log). FA-independence visible: points at the same WR cluster vertically despite different FA (and hence different x-axis positions). Slope is set by WR variation, not FA variation. **C**: sg4(t) for FA $\in \{0.10, 0.20, 0.40, 0.80\}$, WR= 4.8. sg4 continues to grow at $T = 5000$; no saturation reached for FA ≥ 0.20. **D**: Zone-mean $F_z[1]$ (perturbation component) vs. t for all four zones in a representative Exp B seed. Polarity flips visible; dotted lines mark per-zone t_z^*. Inset: distribution of t_z^* by zone across 15 seeds, with verdict "MIXED."

Q2 extended. sg4 does not saturate within $T = 5000$. The saturation timescale exceeds 5000 steps for standard parameters. This is consistent with the amplitude-polarity separation: amplitude (sg4) grows slowly via copy-forward accumulation, while polarity oscillates on a faster timescale.

Q7 resolved. The flip mechanism is a "noisy pitchfork": zones flip collectively (pitchfork-like global instability) but with ~ 600 steps of asynchrony between zones (partial spinodal character). The dominant variability is between seeds (random wave history), not between zones within a seed.

Implications for "critical period." The oscillatory polarity means there is no true "critical period" in the sense of a one-time developmental window. Instead, the VCML zone structure is continuously remixing its polarity. A "critical period" would require suppressing the polarity oscillations – achievable either by increasing crystallization (bad: loses amplitude adaptivity) or by adding a symmetry- breaking mechanism (e.g., asymmetric initial conditions, geographic relay coupling from V99–V100).

8 Conclusion

Paper 27 resolves three open questions from the theory synthesis, with surprising answers:

- **Q1**: $t^* \approx f(\mathrm{WR})$, not $1/(P_c \cdot \mathrm{FA} \cdot \mathrm{WR})$. Perturbation rate controls timing; consolidation rate controls amplitude.
- **Q2**: sg4 does not saturate within $T = 5000$. Amplitude and polarity operate on different timescales.
- **Q7**: Mixed pitchfork–spinodal ($\sigma_{\mathrm{in}}/\sigma_{\mathrm{btw}} = 0.67$). Zone flips are collective but asynchronous.

The core reframing: VCML zone structure is *metastable*, not *committed*. Mandatory turnover drives continuous polarity oscillations that keep the spatial pattern ergodic. This is a direct prediction of the anti-crystallization principle, confirmed at the temporal level.